

Sound Memory-Bounded Quantum Simulation with Abstract Additive Decision Diagrams

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Abstract

Classical simulation is indispensable for developing quantum programs, but explicit state vectors require exponential memory and exact decision diagrams can lose compression on unfavourable circuits. We introduce abstract additive quantum decision diagrams (AAQDDs), a representation that combines shared decision-diagram structure, multiple additive branches, and rectangular complex intervals. A user-supplied per-level node budget triggers sound joins, trading structural memory for precision while allowing simulation to continue. Because correlation-insensitive interval propagation is easily unsound, the implementation separates a useful nominal midpoint from a certificate: for a unitary circuit with exact state ψ and reported midpoint m , it encloses the result using the universal bound $\|\psi - m\|_2 \leq 1 + \|m\|_2$.

We implement the method in the C++23 simulator CORO and evaluate it against Qiskit Aer’s state-vector method and MQT DDSIM. Every one of 48 exact and approximate validation runs contains every reference amplitude. On an eight-qubit adversarial circuit, one setting achieves midpoint fidelity 0.651, while another reduces stored diagram structure from 3.00 to 1.32 MiB. On a deeper 30-qubit adversarial circuit, the bounded modes complete under fixed 10-second and 4-GiB limits where dense and exact decision-diagram baselines do not. The present certificate is deliberately conservative and useful nominal propagation is limited to eight qubits; accordingly, the contribution is a sound memory-bounded mechanism and reproducible reachability result, not a uniformly monotone accuracy curve.

Keywords

quantum circuit simulation, decision diagrams, abstract interpretation, interval arithmetic, approximate computing, reproducibility

1 Introduction

An n -qubit pure state has 2^n complex amplitudes. Dense state-vector simulators are therefore predictable but encounter an exponential memory wall. Decision diagrams instead share repeated substructure and can represent states such as GHZ or separable products in space polynomial in n ; their performance, however, depends strongly on circuit structure and simulation order [1, 7]. Existing approximate decision-diagram techniques commonly discard low-weight contributions while targeting an empirical fidelity [8]. This is valuable, but it does not by itself provide a component-wise sound enclosure of the exact state.

Our starting point is different: when an exact diagram exceeds a memory policy, continue in an abstract domain that contains the exact computation. The idea follows abstract interpretation [2]: replace exact complex coefficients by sets of coefficients, and replace structurally distinct subgraphs by a join. A result then carries

quantified uncertainty rather than silently becoming an unqualified approximation. The engineering difficulty is preserving both soundness and usefulness. Additive diagrams expose correlations between branches; naively joining edge intervals or applying gates to already merged branches can cancel uncertainty and become unsound. Conversely, repeatedly applying global interval matrix bounds is sound but rapidly destroys the midpoint.

This paper makes four contributions:

- We define AAQDDs, whose additive interval-labelled edges represent sets of state vectors.
- We give a memory policy that caps nodes at every nonterminal level and a sound general join for additive contexts.
- We separate nominal propagation from a simple, independently checkable unit-norm certificate, and state precisely where the implementation falls back to a coarse enclosure.
- We provide an open implementation, isolated baseline comparisons, raw data, figures, tests, and one-command build and experiment targets.

The evaluation answers three questions: whether returned rectangles contain exact reference amplitudes (RQ1), whether budgets expose a measurable memory–precision trade-off (RQ2), and whether bounded abstraction reaches circuits that exceed exact methods under identical resource limits (RQ3).

2 Background and Representation

2.1 Quantum states and decision diagrams

Let $\psi \in \mathbb{C}^{2^n}$ be an n -qubit state with $\|\psi\|_2 = 1$. A gate on k qubits is embedded as a unitary transformation U , and circuit simulation computes $\psi_t = U_t \cdots U_1 |0^n\rangle$. A quantum decision diagram recursively decomposes a vector by qubit and shares equal subvectors. Weighted edges factor complex amplitudes; canonical normalization permits extensive sharing on structured states. The same mechanism can nevertheless approach the dense representation on unstructured states.

CORO uses an additive variant. At height $h > 0$, a node contains finite multisets of left and right edges (a_i, D_i) , where each child has height $h - 1$. Its evaluation concatenates two sums:

$$\mathcal{E}_h(L, R) = \left(\sum_{(a,D) \in L} a \mathcal{E}_{h-1}(D) \right), \quad \mathcal{E}_0(\boxed{1}) = 1. \quad (1)$$

Multiple edges to shared children make sums explicit and are useful during gate rewriting.

2.2 Rectangular complex intervals

An abstract coefficient is a closed rectangle $A = [r_-, r_+] + i[s_-, s_+] \subseteq \mathbb{C}$. Interval addition and multiplication outwardly round their endpoints. Replacing every a in Equation 1 by such an A yields a set-valued evaluation $\mathcal{E}^\#(D)$. We order diagrams by concretization:

$D_1 \sqsubseteq D_2$ when $\mathcal{E}^\#(D_1) \subseteq \mathcal{E}^\#(D_2)$ component-wise. A reduction is sound if $D \sqsubseteq R(D)$.

This representation differs from attaching one error scalar to a conventional diagram. Intervals can express asymmetric real and imaginary uncertainty locally, while additive branches retain alternatives created by rewriting. They also expose the central hazard: independently varying occurrences of an interval loses correlations, and later algebraic cancellation cannot be assumed.

3 Memory-Bounded Abstract Simulation

3.1 Budgeted reduction

After each gate, the simulator counts reachable nodes at every nonterminal height. Given a budget $B \geq 1$, it repeatedly selects two nodes from an over-budget level using a minimum-imprecision heuristic and redirects their parents to a merged node. The policy ends when each such level has at most B nodes. B is therefore a structural control, not a direct byte limit: edge-vector capacity and the number of additive branches remain visible in the measured footprint.

For non-additive contexts, one might join corresponding edge coefficients. That construction is not sound in the general additive case because the same child can participate in several sums. Coro instead evaluates a global complex enclosure for each merge operand, joins the two enclosures, and constructs a uniform diagram whose every amplitude is contained by the joined rectangle. This loses precision but is monotone under arbitrary incoming additive coefficients. Parent links are rebuilt before reduction so replacement is scoped to the owned root, and dead branches are removed without deleting shared live children.

3.2 Nominal propagation and certification

A uniform enclosure propagated through many gates using interval row bounds becomes enormous and has zero midpoint. For small diagrams, Coro therefore propagates the reduced structure as a *nominal* interval diagram. That midpoint is useful empirically, but the structural intervals alone are not trusted as a sound certificate after correlation-sensitive rewrites.

Instead, let m be the vector of component-wise rectangle midpoints after nominal propagation. Every supported circuit gate is unitary and the initial state is normalized, so $\|\psi\|_2 = 1$. The triangle inequality gives

$$\|\psi - m\|_2 \leq \|\psi\|_2 + \|m\|_2 = 1 + \|m\|_2 =: \varepsilon. \quad (2)$$

For each component j , both $|\operatorname{Re}(\psi_j - m_j)| \leq \varepsilon$ and $|\operatorname{Im}(\psi_j - m_j)| \leq \varepsilon$. The machine-readable result is therefore the outward-rounded rectangle

$$[\operatorname{Re} m_j - \varepsilon, \operatorname{Re} m_j + \varepsilon] + i[\operatorname{Im} m_j - \varepsilon, \operatorname{Im} m_j + \varepsilon]. \quad (3)$$

Proposition 1. *For a supported unitary circuit, every exact output amplitude belongs to its reported certified rectangle.*

Proof. Equation 2 bounds the Euclidean norm of the error. The magnitude of each component, and hence each of its real and imaginary parts, is no larger than that norm. Outward rounding can only enlarge the resulting rectangle. $\square$

The certificate is universal but weak: even a good midpoint may receive a bound near two. It is intentionally reported separately

from midpoint fidelity. Nominal propagation is limited to eight qubits, based on the measured onset of interval-branch explosion. Once a larger diagram becomes abstract, subsequent gates use a compact uniform unitary-image enclosure. Exhaustive certified output is capped at 16 qubits; structural measurements have no such serialization requirement.

4 Implementation

Coro is a C++23 file-mode and interactive OpenQASM simulator. It supports common one-, two-, and three-qubit unitary gates, parameterized rotations, global phase, and a universal single-qubit gate. OpenQASM 3 provides the surrounding circuit syntax [3]. The directives `@reduce(B)`, `@exact`, `@memory`, and `@intervals` control reduction and machine-readable output.

Structural memory counts each reachable node once and includes allocated edge-vector and parent-vector capacities. It excludes allocator overhead; the comparative harness separately records peak resident set size (RSS) using GNU `time`. The analysis layer transpiles a supported OpenQASM 2 subset, removes final measurements, and rejects classical control rather than changing it silently.

Correctness checks cover interval arithmetic, gate conventions, qubit ordering, shared-node ownership, duplicate additive branches, deterministic reduction properties, enclosure, and large-circuit memory safeguards. The release and ASan/UBSan configurations each execute 121 tests; 32 Python tests cover parsing, transpilation, isolation, resource classification, and Aer/DDSIM integration.

5 Evaluation

5.1 Protocol

Experiments ran on Linux x86-64 with 16 logical CPUs. The checked metadata records Python 3.13.11, Qiskit 2.5.2, Qiskit Aer 0.17.2, and MQT DDSIM 2.5.0. Each comparative configuration runs in a fresh process with a 10-second wall-time limit and 4-GiB address-space limit, repeated three times. Aer uses its state-vector simulator [6]; MQT uses its decision-diagram QASM simulator [5]; both execute one measured shot. Coro runs exact and with $B \in \{1, 4, 16\}$. Successful plots use medians and never turn failures into numeric points.

The 22 circuits comprise scalable separable and GHZ families (10–60 qubits), four-layer deterministic entangling circuits (6–12 qubits), two deep adversarial circuits (20 and 30 qubits, 20 layers), and four small QASMBench kernels [4]. The deep circuits apply seeded $R_x/R_y/R_z$ rotations to every qubit and alternating nearest-neighbour CNOTs. They are explicitly contrived to erode exact decision-diagram sharing.

For RQ1 and RQ2, Aer computes exact vectors for separable, GHZ, layered 6/8-qubit, and four QASMBench instances. Coro runs exact and with $B \in \{1, 2, 4, 8, 16\}$. We reverse state indices to account for Coro’s most-significant-bit convention, then record component containment, normalized midpoint fidelity $|\langle \psi / \|\psi\|, m / \|m\| \rangle|^2$, midpoint error, rectangle diameters, and the certificate of Equation 2.

5.2 Resource scaling and reachability

Figure 1 first provides a sanity check. At 60 qubits, Aer exceeds the memory limit, MQT DDSIM completes in roughly 0.4 s with about 139 MiB RSS, and exact Coro completes in about 0.009 s

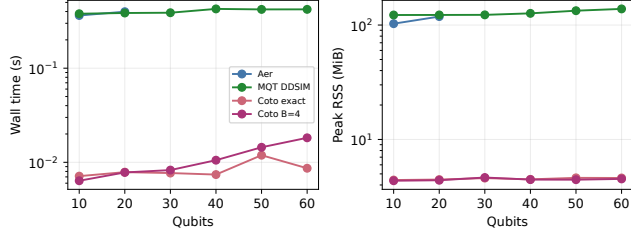


Figure 1: Median whole-process time and peak RSS on separable states. Missing Aer points hit the 4-GiB address-space limit. DD methods compress this favourable family.

Table 1: Deep adversarial circuits under 10 s and 4 GiB. Times are medians of successful runs; failures occurred in all three repetitions.

Qubits	Engine/configuration	Time (s)	Structure
20	Aer state vector	0.652	—
20	MQT DDSIM	timeout	—
20	Coto exact	timeout	—
20	Coto $B = 4$	0.0127	4.42 KiB
30	Aer state vector	memory	—
30	MQT DDSIM	timeout	—
30	Coto exact	timeout	—
30	Coto $B = 4$	0.0270	6.70 KiB

with 12.3 KiB of stored structure. GHZ shows the same qualitative result. These favourable families demonstrate compression but not a unique advantage over existing DD simulation.

The layered cases expose the distinction. Exact Coto times out at 10 and 12 qubits, while all bounded modes complete in milliseconds using only a few KiB after the large-circuit safeguard. The deep 20-qubit case is still tractable for Aer but times out in MQT DDSIM. At 30 qubits, Aer crosses the memory limit and MQT DDSIM and exact Coto time out; bounded Coto completes. Table 1 reports the final medians and statuses.

Thus RQ3 is positive for the deliberately adverse 30-qubit instance: within the stated limits, bounded abstraction returns a sound structural enclosure where both selected alternatives fail. This is a reachability statement, not a claim that it returns a precise state estimate at that scale.

5.3 Soundness and memory-precision behaviour

All 48 validation rows have containment rate 1.0: every exact Aer amplitude lies in the reported rectangle (RQ1). Exact rows have fidelity one and zero certificate radius up to floating-point noise. Approximate rows exercise all five budgets and repeated reductions.

Figure 2 answers RQ2 with an important qualification. On the eight-qubit layered circuit, $B = 4$ gives fidelity 0.651 with certified bound 1.807. $B = 16$ stores 1.32 MiB instead of the exact diagram’s 3.00 MiB, but its fidelity is only 0.174. On the six-qubit layered

circuit, $B = 16$ reaches fidelity 0.628. Several small QASMBench midpoints remain near unit fidelity.

The relationship is not monotone. A smaller node cap can cause a wider merged node to create more additive branches during later gates; budget 1 on the eight-qubit layered case is both larger and slower than several looser settings. Selection is heuristic, so additional nodes also need not improve the final midpoint. The data therefore demonstrate a tunable mechanism with useful operating points, but not an ordered Pareto family. The certified bound remains sound at every point and is kept visually separate from empirical fidelity.

6 Related Work

QMDD-style representations compress quantum vectors and matrices by recursive decomposition and weighted sharing; a broad account and the tool ecosystem are given by Wille et al. [7]. MQT DDSIM is the closest experimental baseline. Simulation-path research shows that contraction order alone can change DD cost by orders of magnitude [1]; our fixed sequential path leaves this optimization complementary.

Zulehner et al. [8] reduce quantum-state DDs using fidelity-oriented schemes and report large empirical savings. AAQDDs instead make uncertainty part of the representation and prioritize set containment. The present midpoint can be compared by the same fidelity metric, but Equation 2 is independent of that comparison. This resembles the sound-overapproximation discipline of abstract interpretation [2], applied to complex amplitudes and graph reduction rather than program fixpoints.

Dense state-vector simulation remains a strong baseline because its cost is regular and its semantics straightforward. Aer also offers several non-dense methods, but we deliberately select its state-vector mode to expose the memory wall and pair it with DDSIM to avoid treating all structural simulation as new. QASMBench supplies externally defined kernels; the adversarial circuits serve a different, clearly labelled purpose by isolating failure modes.

7 Threats to Validity and Limitations

The certificate is mathematically sound but too loose to establish practical accuracy; for an abstract result it is at least one and often near two or more. Deriving a tighter compositional error bound is the main research problem left open. Above eight qubits the safeguarded midpoint is zero, so large reachability results say only that a sound enclosure was computed. Certified amplitude serialization is limited to 16 qubits.

The benchmark uses one measured shot and includes unequal process startup: Aer and DDSIM load Python packages, while Coto is native. The CSV provides Python-engine simulation time, but whole-process comparisons remain imperfect. The 10-second cutoff is intentionally operational, not an asymptotic result. The deep circuits are seeded and reproducible but contrived; real workload distributions may behave differently. Aer has methods beyond state vectors, and MQT offers path choices not evaluated here.

The supported language is unitary-only. Mid-circuit measurement, noise, and classical control are outside scope. Finally, 100% containment on 48 rows plus deterministic property tests is strong validation, not a mechanized end-to-end proof. The repository’s

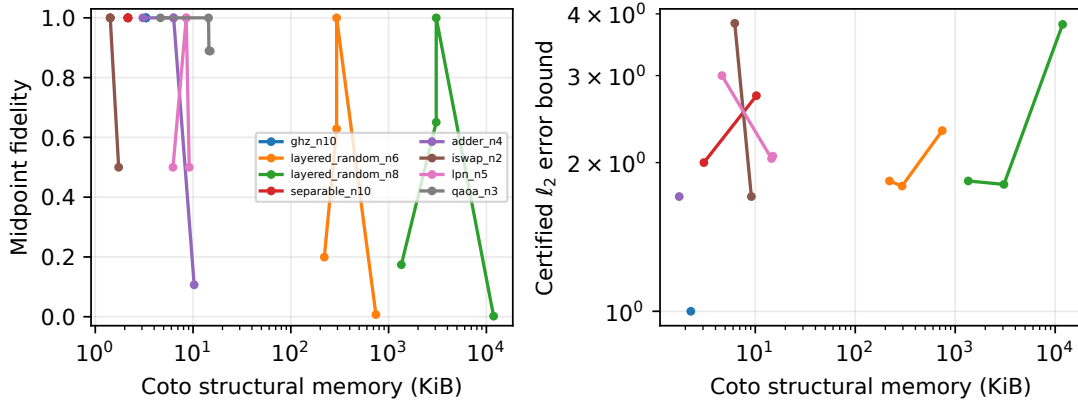


Figure 2: Empirical midpoint fidelity (left) and the independently certified ℓ_2 error bound (right) versus stored Coto structure. Lines connect configurations sorted by memory, not by budget; nonmonotonicity is genuine.

author-review report lists these limitations and open engineering bugs explicitly.

8 Reproducibility

The repository contains source, pinned QASMBench data, raw per-process CSV rows, metadata, plot generation, and the manuscript. The recorded implementation commit is 2bec253; its full hash is in the metadata. From `paper/`, `nix-shell -run 'make pdf'` builds this document with ACM’s `acmart` class. `make verify` builds and tests the simulator; `make experiments` reruns the bounded collectors and figures. Metadata records versions, limits, repetitions, platform, and the implementation commit; failed rows remain in the raw data.

9 Conclusion

AAQDDs turn a fixed structural budget into a sound continuation strategy for quantum simulation. The implementation never equates a high-fidelity midpoint with a guarantee: nominal quality is measured empirically, while a separate unit-norm argument certifies every reported amplitude rectangle. The experiments find useful small-circuit midpoints, expose nonmonotone heuristic behaviour, and show a 30-qubit adverse circuit reachable within fixed limits after both dense and exact-DD alternatives fail. The next step is not a larger claim, but a tighter compositional certificate and a merge policy that makes precision and memory more predictably monotone.

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